

Nate Cirino

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Graduating May 2027 | Available to start May 2027

EDUCATION

Northeastern University

May 2027 | MGPA: 3.6/4.0

B.S. in Computer Science, Systems Concentration

Boston, MA

Coursework: Foundations of Data Science, Algorithms & Data Structures, Object-Oriented Design, Computer Systems, Digital Design & Computer Architecture, Foundations of Software Engineering, Computer Networks, Systems Security, Cybersecurity

EXPERIENCE

Wayfair

Jul 2026 – Present

Software Engineering Co-op

Boston, MA

- Own front-of-house device stack across all Wayfair e-commerce retail brands (AllModern, Birch Lane, Joss & Main, Perigold): e-labels, mobile, and customer/associate-facing POS
- Building a supplier-agnostic e-label API unifying 4 suppliers behind one interface, powering new outlet launches
- Shipped POS checkout flows: in-store scan-to-basket customer switching and live associate-entry customer sync

SculptAI

Jan 2025 – Aug 2026

Software Architect

New York, NY

- Architected Flask backend with retrieval-augmented recommendation pipeline serving personalized workout plans, using Bedrock KB embeddings to rank candidate items per user context
- Evolved retrieval stack from LangChain to Vespa to zero-ops Bedrock KB, lifting recommendation relevance for cold-start and long-tail users while cutting operational overhead
- Sustained 99.99% uptime across 1–2M monthly LLM inference calls for 2K paying users via AWS WAF, Datadog, K8s
- Led eng org restructure at \$1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows; continue as technical advisor on architecture and agentic direction

Wolters Kluwer

Jul 2025 – Jun 2026

Software Engineer

Boston, MA

- Integrated Expert AI, an LLM-backed clinical decision support product, into CME accrual/redemption microservices, expanding GenAI CDS to 250K+ clinicians
- Optimized Expert AI semantic search, retrieval, and filtering across 20+ microservices, improving query relevance and result ranking at production scale
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing workflow
- Overhauled Relay JMS broker serialization with json-io across 20+ microservices, eliminating timeouts at scale

Chevron New Energies

May 2024 – Aug 2024

Software Engineer Intern

Houston, TX

- Built RESTful APIs and data pipelines using Azure Data Factory, improving product efficiency by 40%
- Implemented CI/CD workflows using Azure DevOps and K8s, enabling automated deployment and rollback

PROJECTS

NOAA AUV Sonar Detection | *Python, PyTorch, CNNs, AWS Neuron SDK, SageMaker* April 2025 – July 2025

- Trained deep CNN detection models in PyTorch on thermal sensor spectrograms to identify novel aquatic signatures for NOAA AUVs
- Reformulated architecture and loss to handle severe class imbalance and long-tail sensor noise, improving detection recall on rare signatures
- Designed AWS Neuron SDK deployment path for onboard custom-silicon inference under hard real-time latency limits

Sentinel | *LLM agents, pgvector, FastAPI, Claude, Next.js, OpenTelemetry*

Apr. 2026 – Present

- Built pgvector embedding retrieval over runbooks and postmortems with prompt caching on frozen schemas, reaching 100% eval accuracy on held-out incidents
- Built a self-directed LLM agent that triages incidents via tool use and multi-step reasoning over Prometheus, Tempo, and Loki telemetry
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents

Kepler | *Rust, C++, Raft, LSM-tree, MVCC, Tokio*

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and LSM-tree engine
- Implemented Raft election, log replication, and membership changes; storage: memtable, WAL, leveled SSTables
- Added MVCC transaction layer providing snapshot isolation and linearizable reads across multi-node Raft cluster

SKILLS & HONORS

ML & Frameworks: PyTorch, TensorFlow, Pandas, CNNs, LLMs, RAG & vector retrieval (pgvector, Vespa, Bedrock KB), embeddings, Flask, FastAPI, React, Next.js, Node.js, Express

Languages/Tools: Python, C++, C, Rust, Java, TypeScript, JavaScript, Bash, Linux, AWS (SageMaker, Bedrock, Neuron), Azure, Docker, Kubernetes, PostgreSQL, GraphQL, Terraform, Jenkins, Buildkite, Groovy

AI / Agentic Tooling: Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

Honors: Dean's List (All Semesters), Tech Lead at Northeastern Electronic Racing, PawHacks, json-io OSS Contributor